

Graphwise

Enterprise Semantic Backbone for Precise Context and Efficient AI Reasoning

Executive Summary Report
Generated: October 07, 2026

Executive Summary

Graphwise is an AI Context Platform designed to solve critical challenges in enterprise AI deployment. By leveraging an enterprise semantic backbone—a unified knowledge representation layer—Graphwise enables precise context delivery and deterministic reasoning while maintaining governance, scalability, and accuracy. This report outlines the platform's architecture, maturity model, and comparative advantages in enabling trustworthy, explainable AI applications at enterprise scale.

1. Problem Statement: Hot AI Issues

- **AI is Guessing:** LLMs operate on scattered knowledge without unified understanding, leading to hallucinations and unreliable outputs.
- **Ungoverned Data:** Biased, untraceable outputs and uncontrolled data usage block production deployment.
- **High Costs:** AI operating costs limit scalability, with token consumption and latency being prohibitive factors.
- **Slow Modeling:** Building ontologies has historically taken months, creating deployment delays.

2. The Graphwise Solution: Five Key Innovations

1. **Strict Business Context:** Well-defined vocabulary and aligned data meaning provide the semantic foundation for reasoning.
2. **Deterministic Reasoning:** 100% correct, fast logical inference with complete data validation eliminates guessing.
3. **Precise Context:** Context-rich input (via knowledge graphs) replaces raw text chunks, reducing hallucinations.
4. **AI-Aided Modeling:** Ontologies and taxonomies bootstrapped in minutes using AI, enabling rapid deployment.
5. **Managed Accuracy:** Human-in-the-loop refinement and explainable, grounded answers ensure AI governance.

Key Benefits at a Glance

Capability	Impact
Easier AI Governance	Explainable, grounded answers with definitive retrieval
Token Efficiency	54× less tokens required per query; lower operational costs
Automation	~75% reduction in manual tagging via automated reconciliation
Customer Base	200+ customers, 20+ years of proven semantic platform leadership

3. Graphwise Platform Architecture

The Graphwise platform is structured around a **Content Infrastructure** comprising five key layers:

Layer	Component	Function
Application	Knowledge-centric end-user applications	User-facing workflows embedding AI intelligence
Retrieval	Retrieval and reasoning systems	Retrieve, generate, and reason over information with governance
Knowledge Graphs	Knowledge graphs (ontologies)	Entity and relationship mapping for domain-specific reasoning
Metadata	Semantic metadata (taxonomies)	Standardized definitions ensuring consistent meaning
Data	Multimodal enterprise data	Structured and unstructured data, continuously updated

4. Semantic Layer Maturity: Four Stages

Graphwise defines a maturity progression toward deeper meaning and advanced AI capabilities:

- **Stage 1 – Similarity Search:** Retrieves content by embedding similarity. Best for rapid prototyping.
- **Stage 2 – Metric Consistency:** Encodes meaning in queries and views. Best for analytics and reporting.
- **Stage 3 – Canonical Model:** Defines canonical business objects with consistent identifiers. Best for cross-system use cases.
- **Stage 4 – Semantic Reasoning:** Explicit relationships via ontologies. Best for multi-hop reasoning and compliance.

5. Reasoning Models: Logical vs. Probabilistic

Graphwise employs a hybrid approach, leveraging both deterministic and probabilistic reasoning:

Aspect	Logical Reasoning	Probabilistic (LLMs)
How It Works	Rule-based database reasoner; pre-resolves semantic relationships	LLM chain-of-thought per prompt
Strengths	100% correct, fast, no token cost. Facts reusable.	Handles any logic without modeling. Works on unstructured data.
Limitations	Requires ontology. Structured data only.	Non-deterministic. Slow, expensive. No fact reuse.
Graphwise 80/20	Critical 10–20% of business logic (80% of reasoning work)	Long tail of tasks and common-sense reasoning
Result	Fast, reliable, cost-effective execution	Instant handling with semantic context

6. Token Efficiency: The Measurement

A comparative analysis demonstrates the dramatic token savings achievable by using knowledge graphs instead of raw document context:

Approach	Tokens Required	Efficiency Gain
Context stuffing (all documents)	4,048,224	Baseline
Knowledge graph (evidence only)	74,746	54.2× reduction

Key Finding: Per-question token ranges from 1× (minimal) to 867× (high complexity), demonstrating that knowledge graphs enable precise retrieval of only needed evidence.

7. Emerging Knowledge Formats and Industry Adoption

An ecosystem of semantic technologies is emerging across enterprise platforms:

- **Microsoft Fabric iQ:** No-code ontology authoring with native graph engine.
- **Apache Ossie:** Coalition standard (YAML) for metrics, dimensions, relationships.
- **Databricks Genie:** Continuously-learned graph from data, ranked by authority.
- **Snowflake Cortex Sense:** Hybrid search with governed semantic views.
- **Google Open Knowledge Format:** Knowledge as markdown with YAML frontmatter.
- **Claude's Semantic Layer:** Metric definitions and business context knowledge graphs for analytics.

RDF as Interoperability Foundation: Only RDF offers cross-platform interoperability without expressivity loss.

8. Strategic Implications

For Data Engineers & Knowledge Graph Experts:

- Invest in Stage 4 maturity: ontologies and knowledge graphs are the foundation of trustworthy AI.
- Adopt the Graphwise 80/20 approach: formalize critical business logic via ontologies; use LLMs for the long tail.
- Leverage deterministic reasoning for cost savings and governance. 54× token reduction directly impacts budgets.
- Monitor emerging standards (Ossie, OKF, RDF) to reduce vendor lock-in and enable ecosystem integration.

Enterprise Perspective:

- Enterprise AI governance demands explainability and traceability. Semantic foundations provide both.
- Scalability and cost efficiency are competitive advantages unlocked by knowledge graphs.
- The next generation of AI will compete on precision, not guessing. Semantic context is the differentiator.

Conclusion

Graphwise represents a mature, proven approach to enterprise AI that balances innovation with reliability. By providing precise context through semantic backbones and enabling deterministic reasoning alongside probabilistic AI, Graphwise solves the core challenges limiting AI adoption at scale: cost, accuracy, and governance.

For organizations seeking to move beyond AI experimentation to production deployment, semantic foundations are essential. The choice is no longer whether to invest in knowledge graphs and ontologies, but how quickly to mature the semantic layer.

As enterprise AI matures, semantic precision will be the defining advantage. Organizations that embrace this shift early will lead in trustworthy, scalable, and cost-effective AI delivery.